

Urban Railway Signalling Systems

Wayside, Trackside, OCC & Onboard Equipment Reference

A practical reference to reduce confusion between wayside vs trackside terminology, and to map typical hardware by location and communication path (CBTC / ATC style urban railways).

1. Terminology that causes the confusion

Term	What people usually mean	Better mental model
Wayside	Equipment fixed along the line, not on the train	"Beside the way" = station rooms, cabinets, antennae, interlocking rooms, balises
Trackside	Equipment very close to / on the track	Subset of wayside: axle counters, point machines, track circuits, balises, loops
Station / equipment room	Indoor wayside (often still called wayside)	Zone controllers, interlocking computers, radio base stations
Onboard (trainborne)	Equipment on the train	Vehicle Computer, ATP/ATO, odometry, radio, antennas
OCC / Control Center	Central operations room + servers	ATS/CTC, SCADA, dispatcher HMI, central radio / network

Rule of thumb: Wayside = all fixed ground equipment. Trackside = the outdoor-near-track part of wayside. People saying trackside and wayside interchangeably are often talking about the same ground side, but trackside is more specific.

OCC <--- network/radio ---> Wayside (rooms + trackside) <--- radio/balise/loop ---> Onboard

2. Hardware by location (urban signalling)

Location	Typical equipment / devices	Mainly communicates with	Role (short)
OCC	ATS / CTC servers & workstations	Wayside (ZC/IL), sometimes onboard via radio network	Supervision, routing, schedules, alarms
OCC	SCADA / power & facility HMI	Wayside RTUs, substations, tunnel systems	Power, ventilation, doors, fire
OCC	Central radio / NMS / CCTV / PA	Wayside base stations, cameras, speakers	Comms & situational awareness
Wayside - equipment room	Zone Controller / Wayside ATP computer (CBTC)	OCC, adjacent wayside, onboard (via radio)	Movement authority, train tracking
Wayside - equipment room	Interlocking (CBI / SSI)	OCC, trackside objects, wayside ATP	Points, signals, routes, locking
Wayside - rooms / cabinets	Object controllers / I/O	Interlocking <-> trackside objects	Drive/read points, signals, detectors
Wayside - radio	Radio base stations / APs (Wi-Fi, LTE, TETRA, proprietary)	Onboard radio + backbone to OCC/ZC	Bidirectional CBTC data
Trackside	Point / switch machines	Interlocking via object controller	Move & lock points
Trackside	Signals / indicators (if used; often reduced in CBTC)	Interlocking	Aspect display / fallback
Trackside	Axle counters / track circuits	Interlocking / train detection	Occupancy (fallback or primary in non-CBTC)
Trackside	Balises / beacons / tags	Onboard reader (one-way or limited two-way)	Absolute position, speed limits, mode
Trackside	Inductive loops / waveguide / leaky feeder	Onboard antennas	Continuous / semi-continuous data or radio
Trackside / platform	Platform screen doors (PSD) controllers	Wayside ATP / interlocking / onboard (door enable)	Align & open doors safely

Trackside / platform	Train detection at berth, gap fillers, etc.	Wayside ATP / PSD / interlocking	Precise stopping / door zone
Onboard	Vehicle On-Board Controller (VOBC) / ATP-ATO	Wayside ZC (radio), balises, OCC (via network)	Enforce MA, speed, ATO driving
Onboard	Radio modem / antenna	Wayside radio BS	Continuous CBTC link
Onboard	Balise antenna / interrogator	Trackside balises	Spot localisation / data
Onboard	Odometry (tachometers, IMU, radar, Doppler)	VOBC (local)	Relative position & speed
Onboard	TIU / train interface (brakes, doors, traction)	VOBC <-> train systems	Apply brake, door permit, propulsion cut
Onboard	Driver HMI / DMI	VOBC	Display MA, speed, faults

3. Who talks to whom

From -> To	Typical link	What is exchanged
OCC <-> Wayside rooms	Fibre / IP backbone	Routes, schedules, status, alarms, remote commands
Wayside rooms <-> Trackside objects	Copper / fibre / fieldbus to object controllers	Point position, signal aspect, occupancy, commands
Wayside (ZC) <-> Onboard	Radio (primary in CBTC)	Movement authority, train position reports, health
Trackside balises <-> Onboard	Short-range RF / inductive	Spot data (position ID, gradients, SSR) - usually not via OCC
OCC <-> Onboard	Usually indirect (via wayside radio network / ATS)	Rarely a direct peer link in classic CBTC designs

4. Quick glossary to stop the swap-ups

If someone says...	Ask / interpret as...
Wayside equipment	Anything fixed on ground: rooms and track
Trackside equipment	Only outdoor devices on/beside the rail
Field equipment	Often = trackside objects + cabinets
Central / OCC equipment	ATS, SCADA, dispatchers - not on the line
Trainborne / onboard	Only what rides with the train
Signalling system	All three layers together (OCC + wayside + onboard)

One sentence that helps

Trackside is a place (next to the rails); wayside is a role (ground-fixed signalling); OCC is the brain; onboard is the train.

5. Optional note: CBTC vs conventional fixed-block

In full CBTC, continuous radio between wayside Zone Controller and onboard VOBC carries movement authority; trackside signals and track circuits may be reduced or kept mainly for fallback. In conventional fixed-block, track circuits/axle counters and wayside signals remain primary, and OCC talks mainly to interlocking rather than continuously to each train.

Note: Equipment names vary by supplier (Alstom, Siemens, Thales, Hitachi, CAF, etc.). This document uses generic urban CBTC/ATC roles, not a vendor-specific bill of materials.